

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Regular Examination – Summer 2023 Course: T.Y. B. Tech. Branch : Civil Engineering Semester :VI Subject Code & Name: BTCVPE604D Water Power Engineering Max Marks: 60 Date:19/07/2023 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12M
A)	Define :load factor, capacity factor, utilization factor, diversity factor, Demand factor	CO2	6M
B)	Enlist types of hydropower plant; explain any one in brief..	CO1	6M
C)	A common load is shared by two stations one being a base load plant with 20MW installed capacity and other being a stand by station with 25MW capacity. the yearly output of stand by station is 10×10^6 kwh and that of the base load plant is 120×10^6 kwh. The peak load taken by stand by station is 12MW and this station works for 2500hrs during the year. the base load station takes a peak load of 22.8MW find out: 1) Annual load factor for both stations 2) Plant use factor for both stations 3) Capacity factor for both stations	CO2	6M
Q.2	Solve Any Two of the following.		12M
A)	Define intake, give its type and explain any one.	CO3	6M
B)	Give types of surge tank. Explain any one in detail.	CO2	6M
C)	A penstock with an internal diameter of 1.2 m supplies water at a head equivalent to 17.6 kg/cm². There is a possibility of 20% increase in pressure due to transit condition the design stress and efficiency of the joint may be assume to be 1200kg/cm² and 85% respectively. Calculate the approximately wall thickness of penstock.	CO2	6M
Q. 3	Solve Any One of the following.		12M
A)	With the help of a detailed diagram of power plant outline its basic components and major sections.	CO2	6M

B)	Give types of underground power house .explain any one in detail.	CO2	6M
C)	What are the advantages and disadvantage of underground power house?	CO3	6M
Q.4	Solve Any Two of the following.		12M
A)	Differentiate between impulse and reaction turbine.	CO2	6M
B)	Explain draft tube & give its types. Explain any one in detail.	CO2	6M
C)	Explain with neat sketch governing of impulse turbine.	CO2	6M
Q. 5	Solve Any One of the following.		12M
A)	Give the advantages and disadvantage of tidal power plant.	CO1	6M
B)	Give the advantages and disadvantages of pumped storage plant.	CO3	6M
C)	Explain the concept of pumped storage hydro power plant with its layout	CO3	6M
	*** End ***		